

# Modeling infectious diseases using R

## Practical Session:

### Basic programming in R for the practical sessions

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Sustainable Training in Epidemiological Analysis and Modelling  
for Data-Driven Decision Making in Africa

# What do we cover in this practical session ?

- Basic programming in R.
- Working with objects.
- User functions.
- Working with functions.
- How to produce an R script.
- How to produce an HTML file.

# R programs

- HTML file:
  - eR\_prog0\_STEAM3D\_10\_2026\_V1.html.
- R program:
  - eR-STEAM3D\_Introduction\_V1\_1-\_2026\_R.
- R markdown program:
  - eR\_prog0\_STEAM3D\_10\_2026\_V1.Rmd.

# Objects

- We define two objects x and y (=the data):

```
> x<-c(1.9,1.2,0.7,2.7,1.2,3.1,2.3,2.1,2.1,1.4)
> x
[1] 1.9 1.2 0.7 2.7 1.2 3.1 2.3 2.1 2.1 1.4
> y<-c(1.8,1.1,0.6,2.7,1.4,3.0,2.5,1.9,1.8,1.4)
> y
[1] 1.8 1.1 0.6 2.7 1.4 3.0 2.5 1.9 1.8 1.4
```

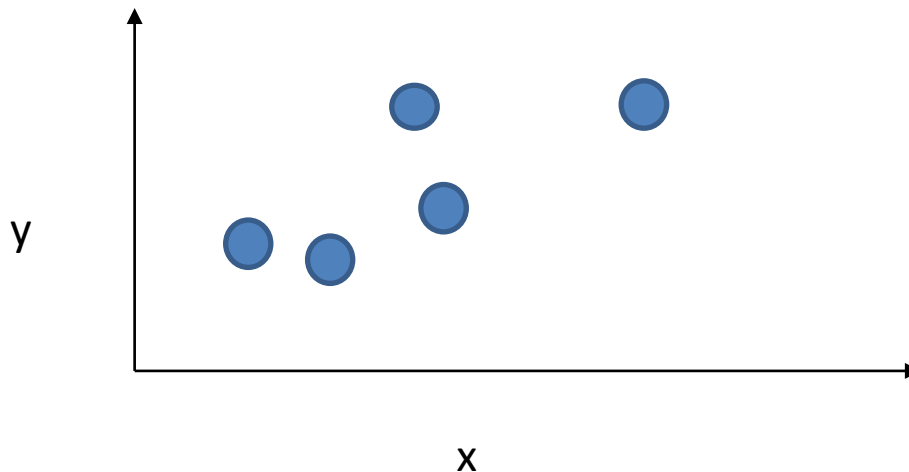
# Data analysis task

1. Calculate the mean of  $x$  and  $y$ .
2. Calculate the correlation between  $x$  and  $y$ .
3. Produce a scatterplot of  $x$  vs.  $y$ .

# How does R work

- R works with functions and objects.
- A function is a code that produce the output of interest.
- For example:
  - To produce a plot we use the function `plot()` .
  - `plot(x,y)` produces the output:

The input  
for the  
function.



# Calculate the mean

- We can calculate the sample mean for x and y using the **mean()** function in R.

**mean**(**data**)



The data.

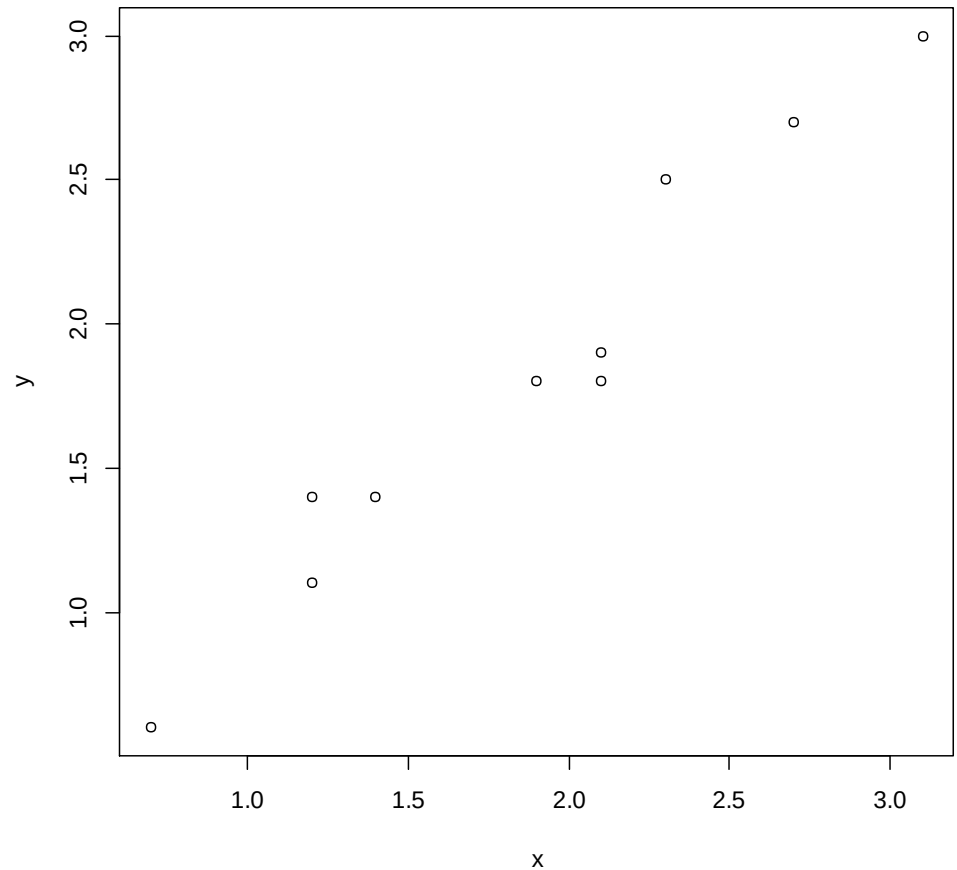
R function to calculate the sample mean.

```
> mean(x)  
[1] 1.87  
> mean(y)  
[1] 1.82
```

# Correlation

```
> cor(x,y)
[1] 0.9774492
```

↓  
R function that calculate  
the correlation.



The function `plot()`  
produces a scatterplot.

```
> plot(x,y)
```

# R functions

- `mean()`, `plot()` ... are function that are a part of the R software.
- We can write our own functions as well.
- Example:
  - Suppose that we want to write a function that
    - Receives two variables as input.
    - Produce the following output:
      - Print the mean of each variable.
      - Print the correlation between the two variables.
      - Produce a scatterplot of the two variables.

# A user function: general form

- A user function is a function that we write to do something in R.

```
function name<-function(x)
```

```
{
```

*R commands (what do you want that the function will do for you.....)*

```
}
```

# A user function: general form

```
function name<-function(x,y)
{  The function starts here

}  The function ends here
```

Input: **x & y**: the sample=the data

Output:

- Descriptive statistics: mean, correlation.
- Graphical output: scatterplot

R functions for the output:

```
mean ()
cor ()
plot ()
```

# A user function: general form

- A small program to produce the output:

```
mean (x)  
mean (y)  
cor (x, y)  
plot (x, y)
```

See slides 5-6.



**x & y**: R objects with the data

# A user function: example

The name of  
the function



```
discript<-function(x,y)
```

```
{
```

```
  print(mean(x))
```

```
  print(mean(y))
```

```
  print(cor(x,y))
```

```
  plot(x,y)
```

```
}
```

The input: R  
objects with  
the data

The function's output:  
calculate the mean and x  
and y, calculate the  
correlation and produce  
a plot.

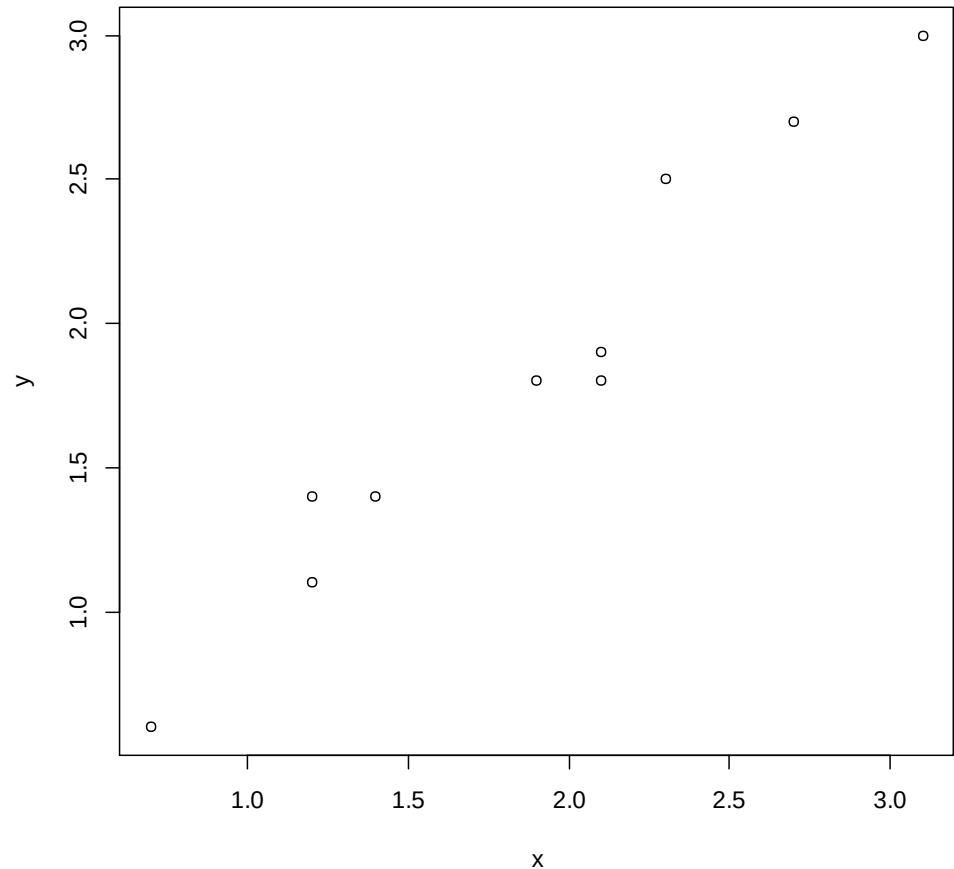


When you finish to write the function, you need to run it first in R.  
Highlight the function code and run.

# Applying the function to the data

```
> discript(x,y)
[1] 1.87
[1] 1.82
[1] 0.9774492
```

- Output:
  - The mean and x and y.
  - The correlation.
  - Scatterplot.



# The cars Data set in R

1. Write **cars** in the script window.
2. Submit

```
> cars
```

```
  speed dist
1      4    2
2      4   10
3      7    4
4      7   22
5      8   16
6      9   10
.      .    .
.      .    .
48     24   93
49     24  120
50     25   85
```

**Data frame in R**

```
> help(cars)
```

```
>
```

## Speed and Stopping Distances of Cars Description

The data give the speed of cars and the distances taken to stop. Note that the data were recorded in the 1920s.

|      |              |         |                        |
|------|--------------|---------|------------------------|
| [,1] | <b>speed</b> | numeric | Speed (mph)            |
| [,2] | <b>dist</b>  | numeric | Stopping distance (ft) |

# What do we want to do ?

- For the cars dataset:
  - Calculate the mean of the cars' speed.
  - Calculate the mean of the cars' stopping distance.
  - Calculate the correlation between cars' speed and stopping distance.
  - Produce a scatterplot.

# Applying the function to the cars data

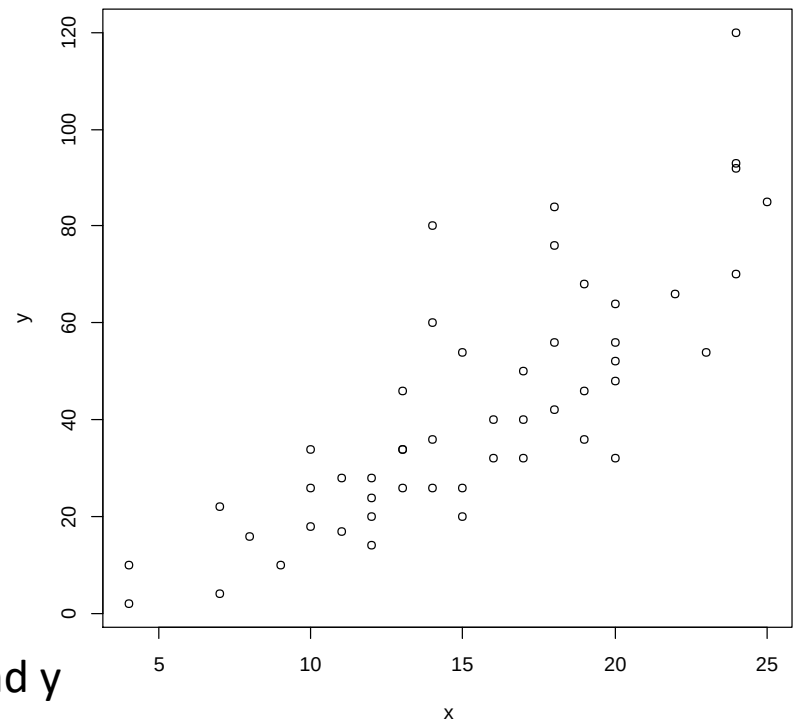
```
> head(cars)
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

The data in R

```
> discript(cars[,1],cars[,2])
[1] 15.4
[1] 42.98
[1] 0.8068949
```

- The function's output:
  - The mean of x.
  - The mean of y.
  - The correlation.
  - Scatterplot.

- The input for the function
- The data = the x and y variables.



How to produce the output presented in the  
slides

# How to produce the output ?

- Open R/R Studio.
- Open the file:
  - [ModelingIDinR0\\_V1\\_Introduction\\_10\\_2026.R.](#)
- Run the code in the file.

# The HTML file

er\_prog0\_STEAM3D\_2026-V1.k

File C:/Ziv\_Temp\_2023/Tarylee\_STEAM-3D/Workshop1\_12\_10\_26/ShortCourse/Practicalessions\_New/er\_prog0\_STEAM3D\_2026-V1.html

uhasseit.be bookmarks

Code

02-10-2026 >eR-BioStat and STAEM-3D

## Modeling infectious diseases using R - basic programming in R

Ziv Shkedy, Rashider Aloni and Leyla Kodalcı

Show

### 1. Defining objects in R

We consider two vectors,  $x$  and  $y$ .

$x = (1.9, 1.2, 0.7, 2.7, 1.2, 3.1, 2.3, 2.1, 2.1, 1.4)$  and  $y = (1.8, 1.1, 0.6, 2.7, 1.4, 3.0, 2.5, 1.9, 1.8, 1.4)$ .

We define the two vectors in R in the following way. For  $x$ :

```
## [1] 1.9 1.2 0.7 2.7 1.2 3.1 2.3 2.1 2.1 1.4
```

and for  $y$ :

```
## [1] 1.8 1.1 0.6 2.7 1.4 3.0 2.5 1.9 1.8 1.4
```

### 2. Calculate the mean of $x$ and $y$ in R

Our first data analysis task is to calculate the mean of  $x$  and  $y$ . In R, the function `mean()` can be used to calculate the sample mean. In our example we use `mean(x)` and `mean(y)`:

```
## [1] 1.87
```

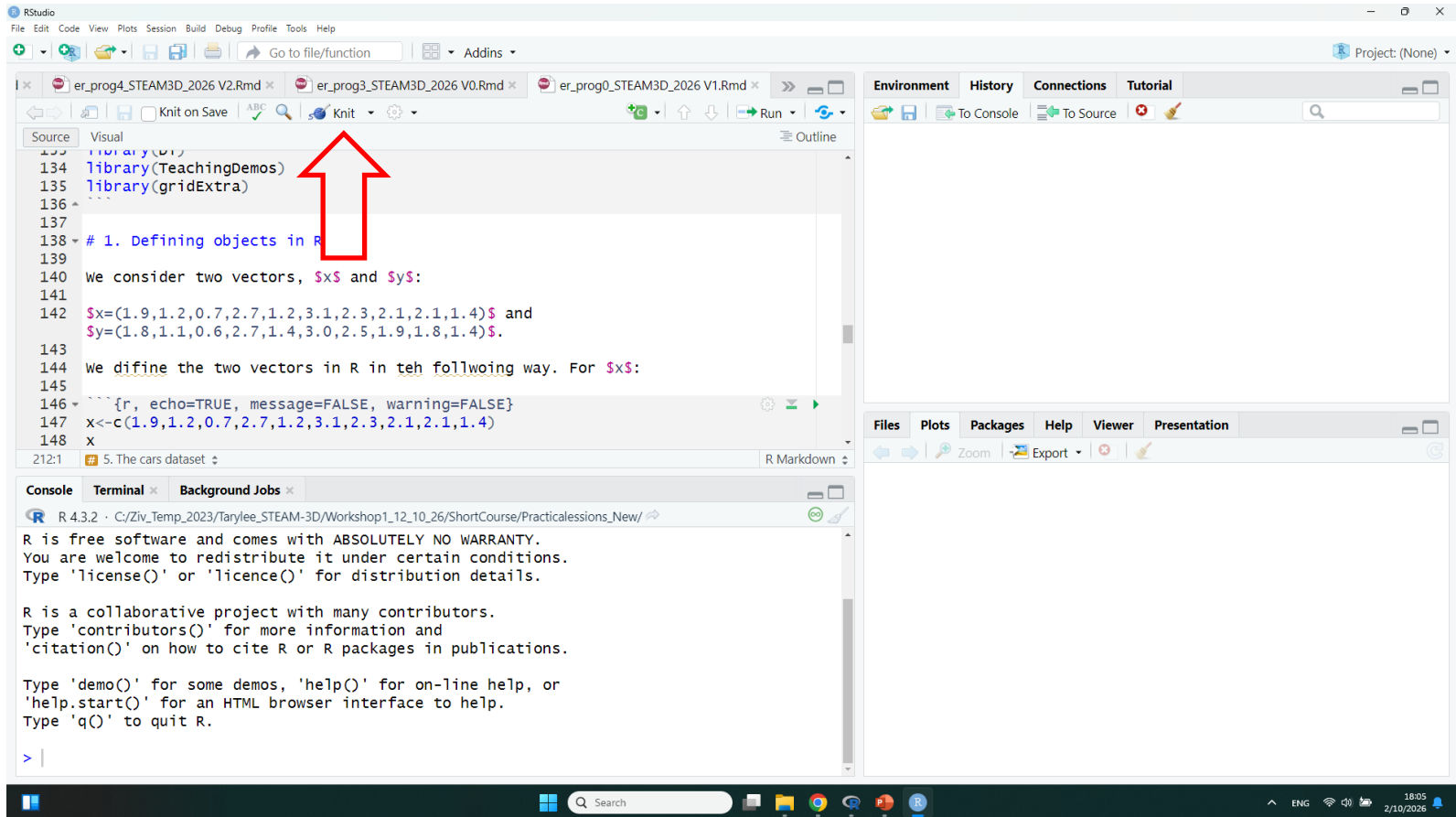
Show

Show

# How to produce the HTML file ?

- Open R studio.
- Open the file:
  - eR\_prog0\_STEAM3D\_10\_2026\_V1.Rmd.
- Knit the file.

# How to produce the HTML file ?



# The R package deSolve

# The `deSolve` package

- We use the R package `deSolve` to run transmission models.
- In particular we use the R function `ode()` to run the model.

# Running a transmission model using the R package `deSolve`

```
require(deSolve)  
out<-as.data.frame(ode(y=state,times=times,func=SIR,parms=parameters))
```



A function in R that can be used for Numerical integration using ordinary differential equation

# Running a transmission model using the R package `deSolve`

```
require(deSolve)  
out<-as.data.frame(ode(y=state,times=times,func=SIR,parms=parameters))
```



- Vectors with input information about the model that we want to run.
- The input that we need to give to the function `ode()`.